D = a - bP + cY *S = d + eP* *D=S requires:* *P* = (a+cY-d)/(e+b)*	**MA Math Econ (630)** **George Mason University** **Fall 2010** **Prof. Roger D. Congleton**	*Maximizing* *U=u(X, Y, Z)* *s.t.* *W = pxX +pyY+pzZ* *requirres* *???*
	Syllabus	
Class Room:	Arlington Campus, room 257	
Day and Time:	Thursday, 7:20 pm - 10:00 pm	
Office Hours:	Tuesday, 1:15-2:45, Thursday 1:15-2:45, and by appointment	
Required Texts:		**SNOW NEWS:**
	Chaing, A. C. and K. Wainwright (2005) *Fundamental Methods of Mathematical Economics* [Paperback] McGraw Hill Higher Education; 4th edition (June 1, 2005)	
	Class Notes for Math Econ: AVAILABLE BELOW (via the web)	
Supplemental Texts		
	McCain, R. A. (2009) *Game Theory and Public Policy* (Edward Elgar - Dec 9, 2009)	
	TENTATIVE COURSE OUTLINE	
Date		**Readings**
9/02	I. Introduction to the Mathematical Analysis of Rational Choice	CW: 1-3, 12.4
	Scope of Course, Usefulness and limitations of deductive methodology	Answers 1
	Models of Rational Choice: net benefits, utility, weak ordering, and transitivity Opportunity Sets and other Constraints. Mathematical concepts: convexity, closed and open sets, compact sets, continuity, functions	
	Economic Application: consumer theory, the theory of the firm	
	Problems V1.1, 1.6, 1.11	
9/09	II. Derivatives and Optimization	CW: 6, 7.1-7.5,12.2-12.7
	Modeling decisions to maximize an objective function: partial derivatives, the chain rule, first and second order conditions, concavity, quasiconcavity, objective function, constraints, optimization using the substitution method	Answers 2
	Economic Applications: the profit maximizing firm, cost-benefit analysis	
	Problems: V3.1, 3.4, 3.5, 3.3, 4.1	
9/16	III. Techniques for Optimization with Equality Constraints	CW: 9, 12, 13.4
	Modeling constrained choices. more applications of the substitution method, an overview of the Lagrangian method. Appendix: the Arrow-Enthoven Sufficiency Theorem	Appendix Answers 3
	Economic Applications: Consumer Theory, Social Welfare Functions	

	Problems: C12.2 1,3,4; C12.5 1,2; V: 7.2, 7.5	
9/23	IV. Comparative Statics: The Implicit Function Theorem	CW: 8.1-8.5 , 13.4-13.6
	How and why changes in constraints affect choices. The Implicit Function Theorem and differentiation rule, the envelop theorem. Applications: how changing prices affect profit maximizing firms and utility maximizing consumers.	Answers 4
	Economic Applications: supply and demand, Cournot reaction functions, comparative statics	
	Problems: C8.5 1,2,3 C8.6 1,2,4 ; V 5.1, 5.2, 5.3, 5.4, 6.1,6.3	
9/30	Help Session on Homework / no lecture / by TA	
10/7	V. Time and Uncertainty in Economic Choices	
	Choices in risky and/or intertemporal settings: Probability Functions, Expected Value, Present Discounted Value, Infinite and Finite Planning Horizons, maximizing expected present values.	Answers 5
	Economic Applications: Intertemporal Choice, Decisionmaking under Uncertainty, Household Finance	
	Problems: C6.6 1,2,3 C6.7 4 C13.5 4,5; V11.5, 11.9, 11.11	
10/14	VI. More on Decision Making under Uncertainty	CW: 13.2-3, 14.1-14.5, 20.4
	Choices in risky and/or intertemporal settings with continuous probabilities and/or time. The use of integrals, integrands, definite and indefinite integrals, risk aversion, subjective rate of time discounting	Answers 6
	Economic Applications: Intertemporal Choice, Decisionmaking under Uncertainty, Continuous Discounting, Calculating Totals from Marginals, Measures of Risk Aversion	
	Problems: C13.2 1,2,4 C13.5 1,2 C13.6 2 ; V11.7; K3:3,4,5,8	
10/14 (supplemental)	VII. Applications of the Rational Choice Model	CW: 16.4-16.5
.	To Crime: The Wealth Maximizing Criminal (Becker, *JPE* 1968: 169-217)	Study Guide 1
.	To Politics: Tax Revenue Maximizing Leviathan (Buchanan and Brennan, *JPubE*, 1977: 255-73)	.
.	To the Selection of a Conservative Central Banker (Waller, C. J., *AER*, 1992: 1006-12)	Answers 7 +
.	To the assignment of liability to lenders: (Lewis and Sappington, *AER*, 2001:724-730.)	.
10/21	VIII. Review for Midterm	2000 Midterm
10/28	***Midterm Exam***	
11/4	IX. General Equilibrium, a Short Overview **(Midterms Returned)**	
	Mathematical Concepts: weak preference ordering, excess demand correspondence, Walras' law. Appendix on sufficient assumptions for the existance of a general equilibrium, Browers fixed point theorem, Kakutani fixed point theorem	
	Economic Applications: Edgeworth Box and Walrasian Equilibrium	Paper Topic Ideas
	Problems: V 17 1-6, 11,13,	

11/11	X. Interdependent Decisions: (1) An Introduction to Game Theory (2) Essential Concepts and Mathematics of Games	M:2-3
	Representing interdependent choices with game matrices: strategy, payoff, non-cooperative games, best reply functions. Named games: prisoner's dilemma game, zero sum games, coordination games, etc.	Answers8
	Economic Applications: Duopoly, Competition, Cartels, Tragedy of the Commons	
	Problems V15.1, 15.3,16.10, V15.2, 15.7	
11/18	XI.More on Non-Cooperative Game Theory	M: 4-7
.	Economic and Political games with strategy continua: duopoly, monopolistic competition, political competition, rent seeking. Technical ideas and terms: Nash equilibrium, pure strategies, mixed strategies, subgame perfect equilibria. Appendix material on the Existance of Nash Equilibrium (Kakutani revisited) Applications from Economics and Politics, interest group politics, majoritarian competition, monopolistic competition	Answers 9
.	Problems M: 13B3, 13B4, 13B8, 14B1, 14B3, 14C4, 14C8	.
11/25	THANKSGIVING BREAK *no class*	
12/2	XII More Applications of Non-Cooperative Game Theory plus Matrix Methods	M: 3-7
	More applications of game theory: signaling games, sorting games, etc. Appendix on matrix algebra, Cramer's rule, derivation of OLS estimator	Study Guide 2
	Applications developed in class, but also if time allows also from: R&D in Duopoly (D'Aspermont and Jacquemin, AER, 1988: 1133-1137.) Theory of Anarchy (Skaperdas, *AER*, 1992:720-739.) Political Influence and Dynamic Consistancy (Garfinkel and Lee, *AER,* 2000: 649-666)	Answers 10 Answers 11
	Problems V15.2, 15.7, M9:B5, B7, B11, B14	
12/09	***6-8 Page Papers Due***	
12/09	XIII. Review for Final	Old PhD Final
12/16	***Final Exam: 7:30-10:15*** (necessarily comprehensive, but oriented toward material covered in second half of the course)	